RISHAB BALASUBRAMANIAN

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About Me

Third-year PhD student in Computer Science at Virginia Tech (advised by Dr. Tu Vu), specializing in language modeling, efficiency, generalization, and post-training alignment. My research focuses on how and why models learn certain behaviors, their structure and transferability, and efficient algorithms for learning them.

Education

- **Virginia Tech** Blacksburg, VA
  *Ph.D. in Computer Science (Advisor: **Tu Vu**)* *Aug 2024 – Jun 2028 (expected)*
- **Oregon State University** Corvallis, OR
  *M.S. in Computer Science & Artificial Intelligence (Advisor: **Huazheng Wang**, **Thesis**)Sep 2021 – May 2024*
- **National Institute of Technology** India
  *B.Tech. in Instrumentation and Control Engineering (**Thesis**)* *Aug 2016 – May 2020*

Publications and Preprints

- ***Spotlight NeurIPS 2026:*** **Rishab Balasubramanian**, Pin-Jie Lin, Rituraj Sharma, Anjie Fang, Fardin Abdi, Viktor Rozgic, Zheng Du, Mohit Bansal, and Tu Vu. The master key hypothesis: Unlocking cross-model capability transfer via linear subspace alignment, 2026
- ***Preprint:*** Salaheddin Alzu'bi, Baran Nama, Arda Kaz, Anushri Eswaran, Weiyuan Chen, Sarvesh Khetan, **Rishab** Bala, Tu Vu, and Sewoong Oh. Roma: Recursive open meta-agent framework for long-horizon multi-agent systems, 2026
- ***EMNLP 2025:*** Pin-Jie Lin, **Rishab Balasubramanian**, Fengyuan Liu, Nikhil Kandpal, and Tu Vu. Efficient model development through fine-tuning transfer, 2025
- ***ICML 2024:*** **Rishab Balasubramanian**, Jiawei Li, Prasad Tadepalli, Huazheng Wang, Qingyun Wu, and Haoyu Zhao. Adversarial attacks on combinatorial multi-armed bandits. In *Forty-first International Conference on Machine Learning*, 2024
- ***TMLR:*** Zichen Wang, **Rishab Balasubramanian**, Hui Yuan, Chenyu Song, Mengdi Wang, and Huazheng Wang. Adversarial attacks on online learning to rank with stochastic click models. *Transactions on Machine Learning Research*, 2024
- ***NeurIPS 2024:*** Zeyu Zhang, Yi Su, Hui Yuan, Yiran Wu, **Rishab Balasubramanian**, Qingyun Wu, Huazheng Wang, and Mengdi Wang. Unified off-policy learning to rank: a reinforcement learning perspective. *Advances in Neural Information Processing Systems*, 36, 2024
- ***Preprint:*** **Rishab Balasubramanian** and Kunal Rathore. Contrastive learning for object detection. *arXiv preprint arXiv:2208.06412*, 2022
- ***IROS 2021:*** **Rishab Balasubramanian**, Lifeng Zhou, Pratap Tokekar, and PB Sujit. Risk-aware submodular optimization for stochastic travelling salesperson problem. In *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 4720–4725. IEEE, 2021
- ***AIAA 2021:*** **Rishab Balasubramanian** and Sujit PB. A cooperative framework for autonomous landings of quadrotors using vision on a moving ugv. In *AIAA Scitech 2021 Forum*, page 1880, 2021

Industry Experience

- **Amazon AGI Customization** — Boston, MA
  *Applied Science II fffffff hosted by **Viktor Rozgic** and **Stefan Stanojevic*** — *2026*
  - Studied the tradeoffs between RL and SFT algorithms, and how to bridge them.
  - Implemented customized task-specific loss functions that minimize forgetting of out-of-distribution (OOD) data, with a $< 1\%$ drop on target tasks.

- **SentientAGI** — Remote
  *Collaboration with **Salaheddin Alzubi** and **Sewoong Oh*** — *2024 – 2025*
  - One of the *first* works to introduce recursive agents.
  - Designed agents, prompts, and evaluation setups for hierarchical problem planning.

- **EndovisionAI** — Remote
  *Research Engineer (**website**)* — *2021*
  - Trained an encoder-decoder model using transformation consistency losses for unsupervised depth estimation from RGB endoscopy images.
  - Generated 3D point clouds from noisy endoscopic images with filtering and estimation.

Research Experience

- **Efficient Learning Algorithms** — Blacksburg, VA
  *Research with **Dr. Tu Vu** (VT)* — *2024 – Present*
  - Working on understanding and transferring capabilities across language model sizes and families.
  - Utilizing model weights and steering vectors for interpretability across models.

- **Adversarial Attacks on Multi-Armed Bandits** — Corvallis, OR
  *Research with **Dr. Huazheng Wang** (OSU)* — *2022 – 2024*
  - Designed novel adversarial attack algorithms on multi-armed bandits.
  - Proposed the *first* algorithm to attack combinatorial multi-armed bandits (CMAB) instances with sublinear cost.

- **Block-wise Trainable Neural Networks** — Remote
  *Research with **Dr. Beidi Chen** (Meta)* — *2022 – 2023*
  - Developed methods to train neural network architectures sequentially to reduce training time, memory consumption, and FLOPs.
  - Enforced gradient blocking and data pruning to further improve efficiency, reducing training time by $\sim 30\%$ while limiting the drop in test accuracy to $< 2\%$.

- **User-Controlled Contrastive Learning for OOD Detection** — Corvallis, OR
  *Research with **Prof. Stefan Lee** (OSU)* — *Mar 2022 – Jun 2022*
  - Following prior on boosting contrastive learning using ranked positives, we proposed a novel method to enforce ranking that leverages user expertise.
  - By incorporating user-defined ranked positives into the contrastive learning framework, we improved performance (accuracy and mAP) on certain tasks, such as object detection and classification, while performance dropped on others, such as OOD detection.
  - We further studied the limitations and explain what might be negatively impacting certain tasks.